Does causal discovery transfer to real measurements? A multi-dataset evaluation on cause–effect pairs, protein signalling and physical devices

Quang-Vinh Dang[1*]

[1*]British University Vietnam, Hung Yen, Vietnam.

Corresponding author(s). E-mail(s): vinh.dq4@buv.edu.vn;

Abstract

Causal discovery methods are usually validated on simulated data. We evaluate fourteen methods on four families of real data with a known or consensus structure: the Tübingen cause–effect pairs, the protein-signalling data of Sachs et al., and two kinds of recordings from the Causal Chambers, physical devices with a known graph. Besides established baselines (PC, GES, DirectLiNGAM, NOTEARS, IGCI, RECI, ANM-HSIC and a pairwise LiNGAM rule), we include six methods we developed for this study and froze before the evaluation: sequential certificates for edges and directions, PC with a nonparametric test, a block-replicated additive method (BTAD), and two classifiers trained on simulated data. The results are mixed and often unfavourable to the new methods. On pairs, a regression-error rule (RECI) is the best or tied best method on every real data set (**61%** on Tübingen, **94%** to **98%** on the chambers), and a classifier that reaches **95%** on simulated pairs is at chance on Tübingen. On graphs, DirectLiNGAM and GES are best on the chamber recordings with independent rows (skeleton $\boldsymbol{F_1}$ **0.55** on the first two, **0.70** on two others), while the methods using nonparametric tests are better on the Sachs data ($\boldsymbol{F_1}$ **0.62** to **0.67** against **0.52** to **0.58**, with **17** reference edges). Seventeen of the nineteen new chamber recordings are random walks with an effective sample size of a few observations out of **4000**, which makes tests that assume independent rows reject roughly five times more graph-implied independences than a block test, and no method exceeds $\boldsymbol{F_1 = 0.42}$ there. We report where each method helps, where it does not, and which of our design choices were fixed before and after seeing the data.

Keywords: causal discovery, benchmark, real data, cause–effect pairs, Causal Chambers, serial dependence

MSC Classification: 62H22 , 62G10 , 68T05

1 Introduction

Methods for learning causal structure from observational data are mostly validated on simulated data, where the generating mechanism is known and the assumptions of the method can be matched to it. Real measurements differ in ways that simulation rarely reproduces: mechanisms are nonlinear and heterogeneous, sensors are collinear, unobserved processes induce dependence between measured variables, and rows are often not independent draws. Benchmarks with real data and a known structure exist, but each covers a narrow slice: the Tübingen cause–effect pairs (Mooij, Peters, Janzing, Zscheischler, & Schölkopf, 2016), the protein-signalling data of Sachs, Perez, Pe'er, Lauffenburger, and Nolan (2005), whose reference network is a contested consensus (Mooij, Magliacane, & Claassen, 2020), and, more recently, the Causal Chambers, physical devices whose graph is known from their construction (Gamella, Peters, & Bühlmann, 2025).

This paper asks a plain question: when methods are applied unchanged to several real data families, which ones work, and do recent ideas beat simple baselines? We evaluate fourteen methods on four families of real data, under a protocol in which every method and every setting was fixed and recorded before the data it is scored on were run, with the exceptions listed in Section 2.4. The methods are eight established baselines (PC, GES, DirectLiNGAM, NOTEARS for graphs; IGCI, RECI, ANM-HSIC and a pairwise LiNGAM rule for pairs) and six methods that we developed during the project that produced this study: sequential certificates for adjacency and direction, PC with a nonparametric test, a block-replicated additive method, and two classifiers trained on simulated data. Including our own methods is deliberate. Several were designed to address failures of the baselines, and the comparison tells whether the design worked on measurements.

Contributions.

1. A multi-dataset evaluation with a recorded protocol. It covers 102 Tübingen pairs, 216 pair analyses of edges of the Causal Chambers, the Sachs data, and 21 chamber recordings with a known graph, and it reports a strict score against the reference graph and a score adjusted for dependences the graph cannot explain.
2. The result that, on pairs, a regression-error rule is the best or tied-best method on every real data set, and a classifier trained on simulated pairs, accurate on its simulated test set, is at chance on Tübingen (Section 4.1).
3. The result that, on graphs, the ranking of methods depends on the data family: DirectLiNGAM and GES on chamber recordings with independent rows, methods with nonparametric tests on the Sachs data, and no method on the random-walk recordings (Section 4.2).
4. A diagnosis of why the walk recordings defeat every method: their exogenous variables are random walks with an effective sample size of a few observations, so tests

that assume independent rows reject graph-implied independences about five times more often than a block test, and a reference set built with them is contaminated (Section 4.3).
5. The places where our own methods do help: control of false edges under repeated inspection and under nonlinear mechanisms in simulation (Section 4.4).

What the paper does not claim.

It does not claim that the methods we did not evaluate (LOCI, QCCD, CAM, DAGMA, PCMCI+ and amortised methods such as AVICI (Lorch, Sussex, Rothfuss, Krause, & Schölkopf, 2022)) fail on these data, and it does not claim that the data sets are representative of causal discovery in general. Several data sets have few reference edges, so most differences between methods are within sampling error, and we give intervals where we can.

2 Data and protocol

2.1 Data

Tübingen pairs.

The 102 scalar-scalar pairs of the cause–effect benchmark (Mooij et al., 2016), with the official weights. Tied values are dequantised by adding uniform noise of the width of the smallest gap, long pairs are subsampled to 2000 rows, and the preprocessing seed is fixed.

Causal Chambers.

The wind tunnel and the light tunnel (Gamella et al., 2025) are physical devices with a graph given by their construction. We use recorded experiments in which the actuators vary, and the nodes of the chamber graph with at least 20 distinct values. Three kinds of recordings occur. In the *i.i.d.* recordings the actuators are redrawn independently at random: the wind tunnel with random intake loads and the light tunnel with uniform inputs, used for the pair and graph evaluations (two recordings with ten subsamples each). In the *walk* recordings the actuators follow random walks: 15 runs of the wind tunnel, the pressure-control experiment, and three light-tunnel experiments, of which one (`color_mix`) is a walk and two have nearly independent rows (Section 4.3). The graphs have 8 to 39 edges among 10 to 20 variables and are bipartite: settings point to measurements.

Sachs.

The 853 observational cells with 11 phosphoproteins, log-transformed; the reference network is the 17-arc consensus of Sachs et al. (2005), which is contested (Mooij et al., 2020).

2.2 Scores

Pairs. Forced-choice accuracy of the direction (unweighted, and weighted by the Tübingen weights), with Clopper–Pearson intervals; and accuracy at a given coverage when pairs are ranked by a method's confidence.

Graphs. Precision, recall and F_1 of the skeleton against the reference graph ("strict"). Because the chamber graphs do not explain all dependence among the sensors (for instance four barometers share an ambient pressure), we also report precision against the graph plus a reference set V of non-adjacent pairs whose graph-implied conditional independence is rejected on held-out rows ("adjusted"); the strict score is the headline.

2.3 Effective sample size

For each recording we measure the lag-1 autocorrelation ρ of the exogenous variables and the implied effective sample size $n(1-\rho)/(1+\rho)$ of the first 4000 rows. The result (Section 4.3) splits the nineteen new recordings into two with an effective sample size above 1000 and seventeen below 20.

2.4 Protocol and provenance

Methods, settings and the scripts that score them were committed to the repository before the run in every case except the following, which are post hoc and flagged where they occur. First, the Causal Chambers recordings with independent rows and the Sachs data were inspected while we were developing the sequential certificates, so those certificates were adjusted after seeing them (a nonparametric test, a larger conditioning order). Second, the nine simulated pair families used to choose the agreement rule of the direction certificate are design data. Third, the reference set V was introduced after seeing false edges. Every other number is from a frozen run: the new chamber recordings, the Tübingen pairs for all methods, the learned models (trained only on simulated data and frozen before evaluation), the baselines, and the Sachs run of all methods. The six methods of the study and their settings are described in Section 3 and the appendix.

3 Methods compared

3.1 Established baselines

Graphs. PC with Fisher-z tests at $\alpha = 0.05$ and 0.01 (Spirtes, Glymour, & Scheines, 2000), GES with the BIC score (Chickering, 2002), DirectLiNGAM (Shimizu et al., 2011) with edges thresholded at $|B| > 0.05$, all from `causal-learn`, and NOTEARS with a linear model (Zheng, Aragam, Ravikumar, & Xing, 2018) (our compact re-implementation, ℓ_1 penalty 0.1, threshold 0.3). We score skeletons only.

Pairs. IGCI in its entropy form with a uniform reference (Daniušis et al., 2010), RECI, which prefers the direction with the smaller regression error after min–max scaling (Blöbaum, Janzing, Washio, Shimizu, & Schölkopf, 2018), an additive noise model with a spline regression and an HSIC independence criterion (ANM-HSIC;

Hoyer, Janzing, Mooij, Peters, and Schölkopf 2009), and the first-order pairwise LiNGAM likelihood-ratio statistic (Hyvärinen & Smith, 2013). These are short re-implementations of the published rules and were not tuned.

3.2 Methods developed for this study

All six were frozen before their evaluation on the data in which they are scored, except where Section 2.4 says otherwise. Details are in the appendix.

Direction certificate (relative fit).

For a pair, two forecasters of the joint law, one per factorisation, are fitted sequentially on past rows; the running mean of the log-score difference has an asymptotic confidence sequence (Waudby-Smith, Arbour, Sinha, Kennedy, & Ramdas, 2024); a direction is reported only when two model classes (constant scale and location-scale generalised Gaussian) agree. It abstains on most pairs. It certifies better predictive fit, not causation.

CERT-CD (sequential adjacency certificates).

An edge is asserted only when, for every conditioning set of size at most k, a sequential test of conditional dependence has rejected. We use the nonparametric version, a sequential generalised covariance test (Shah & Peters, 2020) with spline regressions, at $k = 3$ on the chambers.

PC with a nonparametric test (PC-GCM).

PC (depth at most 3) with the same test used as PC uses a test: cross-fitted residuals, three residual products, an edge is deleted when some conditioning set yields a p-value above α. It separates the effect of the test from the effect of the sequential architecture.

BTAD (block-replicated additive discovery).

Time-ordered rows are cut into 60 contiguous blocks. A variable is added to a node's Markov blanket when its held-out gain in an additive spline model replicates across blocks (one-sided t test, Bonferroni over candidates), blankets are grown and shrunk, and edges that a small subset of the union of two blankets can explain away are removed. Its settings were fixed on simulated data ($\alpha = 0.2$, blanket at most 8).

LRG (learned replicated-gain model).

A gradient-boosting classifier on 25 features per pair (block gains with several conditioning sets, correlation, autocorrelation, within-data-set ranks), trained on 600 simulated data sets with nonlinear mechanisms, serial dependence, latent drift and duplicated sensors. The decision threshold is set on a validation split of the simulations.

PDM (pair-direction model).

A gradient-boosting classifier on 32 features (spline regression errors at three flexibilities, residual dependence, residual shape, IGCI entropies, Gaussian and location-scale

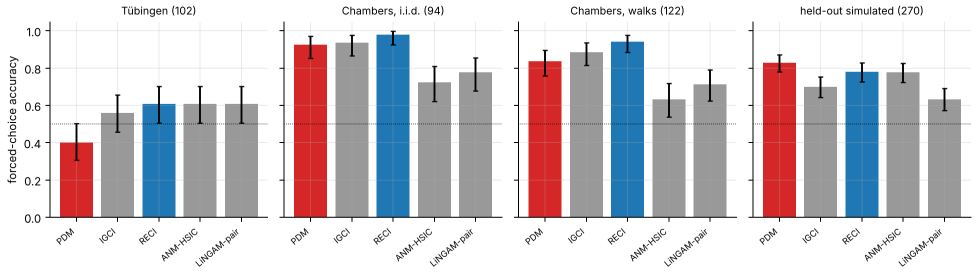


Fig. 1 Forced-choice accuracy with Clopper–Pearson intervals on four pair sets (pairs or pair-runs in parentheses). Red: PDM; blue: RECI; grey: the other rules. Dotted: chance.

log-likelihoods, the pairwise LiNGAM statistic, discreteness) trained on 40000 simulated pairs with 12 mechanism families, 6 noise laws, additive, multiplicative, location-scale and post-nonlinear structure, quantisation and serial dependence. This is the learned approach of Lopez-Paz, Muandet, Schölkopf, and Tolstikhin (2015) with different features and simulator.

4 Results

4.1 Cause–effect pairs

Figure 1 and Table 1 give the forced-choice accuracy of the learned model PDM and the four rules on the four pair sets. RECI is the best or tied-best method on every real set: 60.8% on Tübingen (70.9% weighted), 97.9% on the i.i.d. chamber pairs and 94.3% on the walk pairs. The other rules are lower, and ANM-HSIC and the pairwise LiNGAM rule fall to 63–78% on the chambers. PDM, which reaches 95% on its held-out simulated test set and 83% on nine unseen simulated families, is at chance on the Tübingen pairs (40.2%, interval $[30.6, 50.4]$) and below RECI and IGCI on both chamber sets. The only set on which it is best is the simulated one, which is its own prior. Published methods reach roughly 63–83% forced accuracy on Tübingen (Immer et al., 2023; Mooij et al., 2016; Tagasovska, Vatter, & Chavez-Demoulin, 2020); our RECI is in the lower part of that range.

Abstention.

The direction certificate commits on 12 of the 80 scorable Tübingen pairs and is right on 11. Ranked by its own confidence, RECI is right on 11 of its 12 most confident pairs, and on 21 of 23 against 19 of 23 at the coverage of the constant-scale class. On the chamber pairs the certificate is right on all 31 pairs it commits to (94 pair-runs), as are the RECI and IGCI rankings at that coverage; on the simulated pairs it is right on 78 of 83 against 82 of 83 for RECI. At matched coverage the certificate is therefore not more accurate than RECI; what it adds is that it chooses its own coverage from the data. The Tübingen result of the certificate is stable over random draws of the preprocessing (9/9, 11/11, 10/10 for three further seeds), and its one error occurred on a pair with 254 rows.

Table 1 Forced-choice accuracy (unweighted; weighted by the official weights for Tübingen).

Method	Tübingen (102)	chambers i.i.d. (94)	chambers walks (122)	simulated (270)
PDM	0.402 (0.511)	0.926	0.836	0.830
IGCI	0.559 (0.664)	0.936	0.885	0.700
RECI	0.608 (0.709)	0.979	0.943	0.781
ANM-HSIC	0.608 (0.609)	0.723	0.631	0.778
pairwise LiNGAM	0.608 (0.620)	0.777	0.713	0.633

Table 2 Skeleton precision / recall / F_1 against the reference graph (strict); mean over recordings or subsamples. Sachs: one data set, 17 reference edges. Chambers i.i.d.: two recordings, ten subsamples of 4000 rows each. Chambers with an effective sample size above 1000 (two recordings) and below 20 (seventeen recordings): the first 4000 rows in time order. Best F_1 in bold.

Method	Sachs	Chambers iid	ESS > 1000	ESS < 20
PC, $\alpha = 0.05$	1.00/0.41/0.58	0.40/0.38/0.37	0.48/0.28/0.35	0.36/0.39/0.37
GES-BIC	1.00/0.41/0.58	0.45/0.64/0.49	0.74/0.67/**0.70**	0.26/0.71/0.38
DirectLiNGAM	1.00/0.35/0.52	0.55/0.60/**0.55**	0.71/0.63/0.66	0.26/0.63/0.37
NOTEARS	1.00/0.18/0.30	0.41/0.21/0.27	0.33/0.24/0.26	0.26/0.45/0.33
PC, nonparam. test	0.89/0.47/0.62	0.38/0.33/0.34	0.54/0.34/0.41	0.35/0.54/**0.42**
BTAD	0.89/0.47/0.62	0.56/0.36/0.41	0.82/0.46/0.58	0.60/0.16/0.25
LRG	0.90/0.53/**0.67**	0.39/0.48/0.43	0.52/0.55/0.53	0.51/0.28/0.36
CERT-CD, nonparametric	1.00/0.29/0.45	0.29/0.16/0.20	0.60/0.18/0.28	0.24/0.87/0.37

4.2 Graphs

Table 2 and Figure 2 give the skeleton score on the graph benchmarks. The ranking of methods depends on the data family.

Sachs.

The methods with nonparametric components have the highest F_1 (LRG 0.67, PC with the nonparametric test 0.62–0.64, BTAD 0.62), against 0.58 for PC and GES, 0.52 for DirectLiNGAM and 0.30 for the linear NOTEARS. The reference network has 17 edges, so one edge changes recall by 0.06 and the differences are not statistically distinguishable; the classical methods have the highest precision (1.00) and the new ones find a few more edges.

Chambers with independent rows.

DirectLiNGAM and GES are best. On the two recordings with i.i.d. actuators DirectLiNGAM has $F_1 = 0.55$ and GES 0.49, against 0.32 to 0.43 for PC with either test, BTAD and LRG; on the two further recordings with a large effective sample size

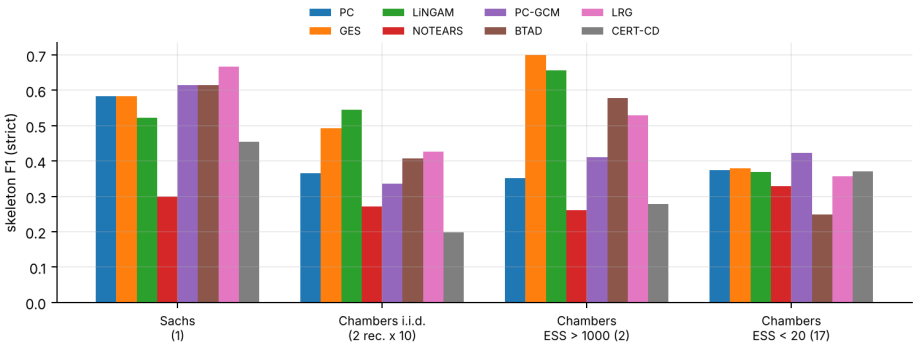


Fig. 2 Strict skeleton F_1 by data family and method.

GES has 0.70 and DirectLiNGAM 0.66. The actuators there are uniform and the mechanisms nearly additive, which suits DirectLiNGAM. BTAD has the highest precision (0.56 and 0.82) at lower recall. The sequential certificate has the lowest F_1.

Random-walk recordings.

On the seventeen recordings with an effective sample size below 20 every method is between 0.25 and 0.42; the best, PC with the nonparametric test, has precision 0.35. BTAD has the highest precision (0.60, with 2.5 false edges per recording against 17 to 51 for PC, GES and DirectLiNGAM) and recall 0.16. Paired comparisons over the 15 wind-tunnel recordings show that BTAD's F_1 is lower than that of every other method (Wilcoxon $p < 0.001$), and that LRG is not distinguishable from PC, GES or DirectLiNGAM.

4.3 The walk recordings carry a few independent observations

Figure 3 shows why the walk recordings defeat every method. In the fifteen wind-tunnel walks the mean lag-1 autocorrelation of the exogenous variables is 0.998 and the implied effective sample size is 3 of 4000 rows (the cross-correlation between a load and the rotation speed it drives is flat for lags from -6 to $+6$, so there is also no usable lag structure). An i.i.d. test of the independences implied by the graph rejects 40.8 non-adjacent pairs per recording; a test on 40 contiguous blocks rejects 8.5. The reference set built with the i.i.d. test, which we used in an earlier analysis, therefore contains many pairs of independent random walks whose dependence changes sign from one recording to the next, and a method scored against it is scored against artefacts. Two recordings of the new set (`actuators_white`, `smooth_polarizers`) have an effective sample size above 1000, and these are the ones on which the classical methods are best.

4.4 Where the new methods help: monitoring and nonlinear mechanisms

On simulated graphs the sequential certificate does what it was designed for (Table 3). When each method is rerun at every sample size from 100 to 6400 on 150 random

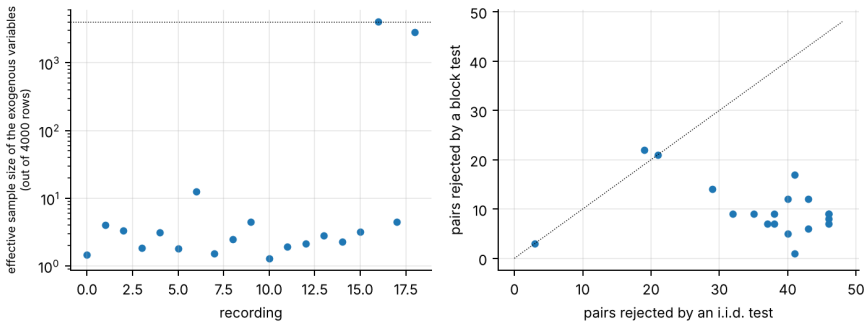


Fig. 3 Left: effective sample size (log scale) of the exogenous variables in each of the nineteen new recordings. Right: number of non-adjacent pairs whose graph-implied independence is rejected by an i.i.d. test and by a block test.

Table 3 Probability that a false edge is ever reported over $n = 100, \ldots, 6400$ (150 graphs, $d = 8$), and graphs with a false edge under nonlinear mechanisms (60 graphs).

Method	Laplace errors	Gaussian errors
CERT-CD, linear test	0.027	0.040
CERT-CD, nonparametric test	0.060	0.047
PC, $\alpha = 0.05$ / 0.01	0.747 / 0.267	0.773 / 0.320
PC, nonparametric test, $\alpha = 0.05$ / 0.01	0.767 / 0.387	0.787 / 0.320
GES-BIC	0.913	0.960
DirectLiNGAM	0.760	1.000
nonlinear mechanisms: linear test (of 60)	33	28
nonlinear mechanisms: nonparametric test (of 60)	1	2

eight-node graphs and a graph counts as a failure if any estimate contains a false edge, CERT-CD fails in 2.7% of the graphs under Laplace errors, against 27% to 91% for PC, GES and DirectLiNGAM and 39% to 77% for PC with the same nonparametric test, so the effect is due to the delete-on-non-rejection architecture and not to the test. With nonlinear mechanisms the linear test of the certificate fails in 33 of 60 graphs under Laplace errors and 28 of 60 under Gaussian errors, and the nonparametric version in 1 and 2 of 60. The guarantee of the linear version is exact only for Gaussian regression errors, and that of the nonparametric version is asymptotic. These are simulations within the models; on the chambers the false-edge control did not hold (a false edge adjusted for a reference set built with an i.i.d. test in 12 of 19 recordings, a strict false edge in 18).

4.5 The gap between simulation and measurement

The two learned models are accurate on data from their training distribution and lose that on real data (Table 4). PDM falls from 0.95 to 0.40 on Tübingen; LRG falls

Table 4 Learned models: simulated test against real data.

Model	simulated test	real data	measure
PDM	0.95 / 0.83	0.40 Tübingen; 0.93, 0.84 chambers	forced accuracy
LRG	0.86	0.67 Sachs; 0.43, 0.53, 0.36 chambers	skeleton F_1

from an F_1 of 0.86 on simulated graphs to 0.36–0.43 on the chambers, and its within-recording ranking of pairs on the walk recordings is close to chance (mean AUC 0.55). We cannot say which feature of the real data is responsible; the simulators contain serial dependence, latent drift, duplicated sensors and several mechanism families and still do not cover them.

5 Discussion

What the evidence supports.

On the real data we examined, simple methods are hard to beat. On pairs the best method is RECI, which compares two regression errors; on graphs the best methods on recordings with independent rows are DirectLiNGAM and GES, and on the Sachs data a mix of methods that use nonparametric tests, with differences smaller than the sampling error of a 17-edge reference. The methods we developed to address specific failures show their intended benefit in simulation and some of it on measurements: lower false-edge rates under repeated inspection and nonlinear mechanisms in simulation, and the highest precision on the chamber recordings, BTAD, at the price of recall. None of them has the best F_1 on the chambers. The two classifiers trained on simulated data are the clearest negative results, and we think they are informative: a model that is nearly perfect on a broad simulator can be at chance on real pairs.

Implications for evaluation.

First, the effective sample size of a recording should be reported: seventeen of nineteen new chamber recordings had fewer than twenty effective observations, which no test that assumes independent rows can cope with, and which also contaminates reference sets built with such tests. Second, adjusting the score for dependences the reference graph cannot explain changes conclusions and should be shown next to the strict score, with the construction of the adjustment disclosed and made independent of the methods. Third, the protocol matters: several of our early positive results were produced by choices made after seeing the data, and the frozen evaluation reversed them. We report which numbers are frozen and which are not (Section 2.4).

Limitations.

The reference networks are small, the Sachs consensus is contested, and the chamber graphs are bipartite, so recovery of the skeleton is mostly feature selection from actuators to measurements. Most differences between graph methods are within sampling error, and the effective sample size of the walk recordings limits what any

method can show. We did not evaluate the amortised methods, CAM, DAGMA, LOCI, QCCD or PCMCI+, and our baselines are re-implementations with default settings; a better-tuned baseline could change a ranking. The Causal Chambers also contain interventional experiments, where a simple shift test already recovers the descendants with perfect precision, and we did not use them. The new methods were developed by the author of this paper, which is a source of bias; the frozen protocol and the public code are the mitigation.

Competing interests. The author declares no competing interests.

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Ethics approval. Not applicable. The study involves no human participants or animals. The protein-signalling data analysed in Section 2 are publicly available and were published by Sachs et al. (2005).

Data and code availability. All simulation code, experiment scripts, random seeds and figure-generating scripts are available at https://github.com/vinhqdang/SPRINT-CD. The protein-signalling data of Sachs et al. (2005) are redistributed with the `bnlearn` package (Scutari, 2010) and are downloaded by the experiment script rather than included in the repository. The Tübingen cause-effect pairs of Section 2 are downloaded on first use from https://webdav.tuebingen.mpg.de/cause-effect/ and are not redistributed.

Author contributions. The author is solely responsible for all aspects of this work.

Use of generative AI. A large language model was used in support of writing this manuscript. The author verified and modified the resulting content and takes full responsibility for the manuscript in its entirety, including its claims, derivations, experiments and references.

Appendix A Method details

Pair features.

The direction hypothesis $c \to e$ is described by the log cross-validated mean squared error of additive cubic spline regressions of e on c with 2, 4 and 8 knots, the log HSIC between the residual and the cause, the share of $\log r^2$ explained by c, the skewness and kurtosis of the residual, the correlation of r^2 with c^2, held-out Gaussian and location-scale log-likelihoods, and the spacing entropy, skewness and kurtosis of c; the feature vector is the difference between the two directions plus their sum and symmetric nuisance features (sample size, correlation, share of distinct values). The classifier has depth 6, 500 trees and learning rate 0.05.

Graph features (LRG).

For every pair, the held-out gain of adding one variable to a model of the other with 40 contiguous blocks (mean, t statistic, share of positive blocks), in both directions, with an empty conditioning set, the three strongest other predictors, and the single

conditioner that most explains the pair away; the absolute correlation; the smaller and larger lag-1 autocorrelation of the two variables; the log number of variables; and within-data-set ranks of the three t statistics. Depth 4, 300 trees. The decision threshold (0.4) maximises validation F_1.

BTAD.

Spline ridge models with three knots at the quartiles, inputs clipped at the 1% and 99% quantiles, pairwise products when the conditioning set has two or more members; gap of 5% of the block length between blocks; 60 blocks; Bonferroni over the candidates of each step; blanket of at most 8 variables; spouse removal over subsets of size at most 2 of the union of the blankets.

PC with a nonparametric test.

Residuals of X_v and X_v^2 on an additive spline of X_S with two-fold cross-fitting; the three products $r_i r_j$, $r_i r_j^{(2)}$ and $r_j r_i^{(2)}$; p-value $= \min(1, 3 \min_k p_k)$.

Sequential certificates.

The adjacency certificate and the direction certificate are described, with their assumptions, in the companion paper (`paper/certcd2` in the repository); here only their outputs are used.

Reproduction.

Each experiment is a script in `experiments/`, committed before it was run, with a summary script in `scripts/`; the figures are made by `scripts/make_benchmark_figs.py`. All results are in `results/` as JSON. Seeds are fixed. The Chamber data and the Tübingen pairs are downloaded on first use.

References

Blöbaum, P., Janzing, D., Washio, T., Shimizu, S., Schölkopf, B. (2018). Cause-effect inference by comparing regression errors. *Proceedings of the 21st international conference on artificial intelligence and statistics.*

Chickering, D.M. (2002). Optimal structure identification with greedy search. *Journal of Machine Learning Research*, *3*, 507–554,

Daniušis, P., Janzing, D., Mooij, J., Zscheischler, J., Steudel, B., Zhang, K., Schölkopf, B. (2010). Inferring deterministic causal relations. *Proceedings of the 26th conference on uncertainty in artificial intelligence.*

Gamella, J.L., Peters, J., Bühlmann, P. (2025). Causal chambers as a real-world physical testbed for AI methodology. *Nature Machine Intelligence*, , https://doi.org/10.1038/s42256-024-00964-x

Hoyer, P.O., Janzing, D., Mooij, J.M., Peters, J., Schölkopf, B. (2009). Nonlinear causal discovery with additive noise models. *Advances in neural information processing systems 21* (pp. 689–696).

Hyvärinen, A., & Smith, S.M. (2013). Pairwise likelihood ratios for estimation of non-Gaussian structural equation models. *Journal of Machine Learning Research*, *14*, 111–152,

Immer, A., Schultheiss, C., Vogt, J.E., Schölkopf, B., Bühlmann, P., Marx, A. (2023). On the identifiability and estimation of causal location-scale noise models. *Proceedings of the 40th international conference on machine learning.*

Lopez-Paz, D., Muandet, K., Schölkopf, B., Tolstikhin, I. (2015). Towards a learning theory of cause-effect inference. *Proceedings of the 32nd international conference on machine learning.*

Lorch, L., Sussex, S., Rothfuss, J., Krause, A., Schölkopf, B. (2022). Amortized inference for causal structure learning. *Advances in neural information processing systems* (Vol. 35).

Mooij, J.M., Magliacane, S., Claassen, T. (2020). Joint causal inference from multiple contexts. *Journal of Machine Learning Research*, *21*(99), 1–108,

Mooij, J.M., Peters, J., Janzing, D., Zscheischler, J., Schölkopf, B. (2016). Distinguishing cause from effect using observational data: Methods and benchmarks. *Journal of Machine Learning Research*, *17*(32), 1–102,

Sachs, K., Perez, O., Pe'er, D., Lauffenburger, D.A., Nolan, G.P. (2005). Causal protein-signaling networks derived from multiparameter single-cell data. *Science*, *308*(5721), 523–529, https://doi.org/10.1126/science.1105809

Scutari, M. (2010). Learning Bayesian networks with the bnlearn R package. *Journal of Statistical Software*, *35*(3), 1–22, https://doi.org/10.18637/jss.v035.i03

Shah, R.D., & Peters, J. (2020). The hardness of conditional independence testing and the generalised covariance measure. *The Annals of Statistics*, *48*(3), 1514–1538, https://doi.org/10.1214/19-AOS1857

Shimizu, S., Inazumi, T., Sogawa, Y., Hyvärinen, A., Kawahara, Y., Washio, T., ... Bollen, K. (2011). DirectLiNGAM: A direct method for learning a linear non-Gaussian structural equation model. *Journal of Machine Learning Research*,

12, 1225–1248,

Spirtes, P., Glymour, C., Scheines, R. (2000). *Causation, prediction, and search* (2nd ed.). Cambridge, MA: MIT Press.

Tagasovska, N., Vatter, T., Chavez-Demoulin, V. (2020). Distinguishing cause from effect using quantiles: bivariate quantile causal discovery. *Proceedings of the 37th international conference on machine learning.*

Waudby-Smith, I., Arbour, D., Sinha, R., Kennedy, E.H., Ramdas, A. (2024). Time-uniform central limit theory and asymptotic confidence sequences. *The Annals of Statistics*, *52*(6), 2613–2640,

Zheng, X., Aragam, B., Ravikumar, P., Xing, E.P. (2018). DAGs with NO TEARS: continuous optimization for structure learning. *Advances in neural information processing systems* (Vol. 31).